

[Before beginning, rename column `product_key` in `sales` table to '`s_product_key`', and that in `products` table to '`p_product_key`']

1. Create a JOIN as '`sales LEFT JOIN products`' & return its shape. Expected output as formatted ahead:

Info	Value
# Records	60,398
# Columns	20

2. From table '`products`', select column '`product_name`' & trim the leading/trailing spaces around the hyphen in values. Expected output:

product_name	cleaned_product_name
HL Road Frame - Black- 58	HL Road Frame-Black-58
HL Road Frame - Red- 58	HL Road Frame-Red-58
Mountain-100 Black- 38	Mountain-100 Black-38
Mountain-100 Black- 42	Mountain-100 Black-42
Mountain-100 Black- 44	Mountain-100 Black-44

3. In table '`sales`', shift column '`price`' to the right of column '`due_date`', and then column '`quantity`' to right of column '`price`'. Expected output: ↓

order_number	s_product_key	customer_key	order_date	shipping_date	due_date	price	quantity	sales_amount
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4. From '`products`' table, fetch details of all red-colored products ('Taillights - Battery-Powered' should NOT be listed despite the 'red' in 'Powered').

product_name
LL Road Frame - Red- 58
LL Road Frame - Red- 60
LL Road Frame - Red- 62
Taillights - Battery-Powered
Road-250 Red- 58
HL Road Frame - Red- 44
HL Road Frame - Red- 48

Wrong output

product_name
LL Road Frame - Red- 52
LL Road Frame - Red- 58
LL Road Frame - Red- 60
LL Road Frame - Red- 62
Road-250 Red- 58
HL Road Frame - Red- 44
HL Road Frame - Red- 48

Correct output

5. Count the number of NULLs (or 'NA') in each column of the '`products`' table. Expected output in descending order of NULLs count:

Column_Header	NULLS_Count
product_line	17
category	7
subcategory	7
maintenance	7
cost	2
p_product_key	0
product_id	0
product_number	0
product_name	0
category_id	0
start_date	0

6. Fetch those dates of the calendar on which absolutely zero sales happened.

zero_sales_dates
2011-02-13
2011-04-23
2011-03-11

7. Which category hasn't been classified into any of the product lines?
 8. Find the range of sales (i.e. min. sales amount and max. sales amount) for each sub-category ordered during 2013.
 9. Calculate the rolling average sales of categories spread over 3-months window. **Random snapshot** of expected output ahead:

category	Month	sales	prev_3_months_avg_sales
Accessories	10	66103	61308.00
Accessories	11	67198	63922.33
Accessories	12	67863	67054.67
Bikes	1	1795001	1795001.00
Bikes	2	1677034	1736017.50
Bikes	3	1826336	1766123.67
Bikes	4	1871194	1791521.33
Bikes	5	2120578	1939369.33
Bikes	6	2844032	2278601.33
Bikes	7	2325334	2429981.33
Bikes	8	2595180	2588182.00
Bikes	9	2450441	2456985.00
Bikes	10	2816603	2620741.33
Bikes	11	2881799	2716281.00
Bikes	12	3108125	2935509.00
Clothing	1	24033	24033.00
Clothing	2	21281	22657.00
Clothing	3	26759	24024.33
Clothing	4	25371	24470.33

10. How many mountain bikes and road bikes were sold from Apr 2012 till Mar 2013?

11. Calculate the cumulative sales of products over the months of 2011 and 2012. Here is a **random snapshot** of the desired output:

order_year	product_number	product_name	Month	sales	cumulative_sales
2012	BK-M68S-46	Mountain-200 Silver- 46	9	49704	221597
2012	BK-M68S-46	Mountain-200 Silver- 46	10	55917	277514
2012	BK-M68S-46	Mountain-200 Silver- 46	11	37278	314792
2012	BK-M68S-46	Mountain-200 Silver- 46	12	62877	377669
2012	BK-M38S-40	Mountain-400-W Silver- 40	12	769	769
2012	BK-M18B-40	Mountain-500 Black- 40	12	540	540
2012	BK-M18S-44	Mountain-500 Silver- 44	12	565	565
2012	PK-7098	Patch Kit/8 Patches	12	8	8
2012	SO-R809-L	Racing Socks- L	12	18	18
2012	BC-R205	Road Bottle Cage	12	81	81
2012	TT-R982	Road Tire Tube	12	28	28
2012	BK-R89B-44	Road-250 Black- 44	1	32730	32730
2012	BK-R89B-44	Road-250 Black- 44	2	30548	63278
2012	BK-R89B-44	Road-250 Black- 44	3	19638	82916
2012	BK-R89B-44	Road-250 Black- 44	4	17456	100372
2012	BK-R89B-44	Road-250 Black- 44	5	19638	120010
2012	BK-R89B-44	Road-250 Black- 44	6	17456	137466
2012	BK-R89B-44	Road-250 Black- 44	7	28366	165832
2012	BK-R89B-44	Road-250 Black- 44	8	26184	192016

12. Which red-colored products were never sold? Expected output (has 22 records):

product_id	product_number	product_name	order_date	sales_amount
211	FR-R92R-58	HL Road Frame - Red- 58	NULL	NULL
240	FR-R92R-62	HL Road Frame - Red- 62	NULL	NULL
243	FR-R92R-44	HL Road Frame - Red- 44	NULL	NULL
246	FR-R92R-48	HL Road Frame - Red- 48	NULL	NULL
249	FR-R92R-52	HL Road Frame - Red- 52	NULL	NULL
252	FR-R92R-56	HL Road Frame - Red- 56	NULL	NULL
263	FR-R38R-44	LL Road Frame - Red- 44	NULL	NULL
265	FR-R38R-48	LL Road Frame - Red- 48	NULL	NULL
267	FR-R38R-52	LL Road Frame - Red- 52	NULL	NULL
269	FR-R38R-58	LL Road Frame - Red- 58	NULL	NULL
271	FR-R38R-60	LL Road Frame - Red- 60	NULL	NULL
273	FR-R38R-62	LL Road Frame - Red- 62	NULL	NULL
274	FR-R72R-44	ML Road Frame - Red- 44	NULL	NULL
275	FR-R72R-48	ML Road Frame - Red- 48	NULL	NULL
276	FR-R72R-52	ML Road Frame - Red- 52	NULL	NULL
277	FR-R72R-58	ML Road Frame - Red- 58	NULL	NULL
278	FR-R72R-60	ML Road Frame - Red- 60	NULL	NULL
315	BK-R68R-58	Road-450 Red- 58	NULL	NULL
316	BK-R68R-60	Road-450 Red- 60	NULL	NULL
317	BK-R68R-44	Road-450 Red- 44	NULL	NULL
318	BK-R68R-48	Road-450 Red- 48	NULL	NULL
319	BK-R68R-52	Road-450 Red- 52	NULL	NULL

13. List the subcategories of each category. Expected output as formatted ahead:

Categories	Subcategories
Accessories	Bike Racks, Bike Stands, Bottles and Cages, Cleaners, Fenders, Helmets, Hydration Packs, Lights, Locks, Panniers, Pumps, Tires and Tubes
Bikes	Mountain Bikes, Road Bikes, Touring Bikes
Clothing	Bib-Shorts, Caps, Gloves, Jerseys, Shorts, Socks, Tights, Vests
Components	Bottom Brackets, Brakes, Chains, Cranksets, Derailleurs, Forks, Handlebars, Headsets, Mountain Frames, Road Frames, Saddles, Touring Frames, Wheels

14. Calculate the percentage contribution of each category's sales to overall sales. Expected output as formatted ahead:

category	sales	overall_sales	%_of_overall_sales
Bikes	28,316,272.00	29,355,985.00	96.5%
Accessories	699,997.00	29,355,985.00	2.4%
Clothing	339,716.00	29,355,985.00	1.2%
Components	0.00	29,355,985.00	0.0%

15. Find the average **price** for each **product line** for the year 2012. Expected output as formatted ahead:

product_line	line_2012_avg_price
Road	1625.66
Mountain	2015.77
Touring	1262.24
Other Sales	24.89

16. Find the lowest revenue generating **subcategory** for each **category** during the last quarter of **2013**. Expected output as ahead:

category	subcategory
Accessories	Cleaners
Bikes	Touring Bikes
Clothing	Socks

17. What was the **net sales** on **weekends** during the years **2011, 2012** and **2013**? Expected output as formatted ahead:

order_year	net_weekend_sales
2011	2,063,746.00
2012	1,743,293.00
2013	4,416,966.00

18. Fetch each sub-category's annual sales, previous year's sales, sales difference from previous year, sales change factor. Expected output as formatted:

year	subcategory	sales_annual	sales_prev_year	sales_diff	sales_growth_factor
2010	Mountain Bikes	16,975.00	NULL	NULL	NULL
2011	Mountain Bikes	1,349,343.00	16,975.00	1,332,368.00	78.5
2012	Mountain Bikes	3,612,491.00	1,349,343.00	2,263,148.00	1.7
2013	Mountain Bikes	9,947,639.00	3,612,491.00	6,335,148.00	1.8
2010	Road Bikes	26,444.00	NULL	NULL	NULL
2011	Road Bikes	5,769,164.00	26,444.00	5,742,720.00	217.2
2012	Road Bikes	9,324,069.00	5,769,164.00	3,554,905.00	0.6
2013	Road Bikes	14,519,438.00	9,324,069.00	5,195,369.00	0.6
2012	Touring Bikes	21,390.00	NULL	NULL	NULL
2013	Touring Bikes	3,844,580.00	21,390.00	3,823,190.00	178.7

19. How many **categories** do NOT belong to any **product line**?

20. Fetch the details of the products that remained unsold over the years (165 records expected in output).

21. Which subcategories remained unsold between January 01, 2011 and December 31, 2013? Expected output:

# Unsold	Unsold SubCategories
19	Bib-Shorts, Bottom Brackets, Brakes, Chains, Cranksets, Derailleurs, Forks, Handlebars, Headsets, Lights, Locks, Mountain Frames, Panniers, Pumps, Road Frames, Saddles, Tights, Touring Frames, Wheels

22. Find the sales of S/M/L/XL **sleeved** clothing over the 2012, 2013, & 2014. Expected output as formatted ahead (any permutation of these records):

Product	2012_Sales	2013_Sales	2014_Sales
Long-Sleeve Logo Jersey- M	50.00	20,850.00	1,200.00
Long-Sleeve Logo Jersey- L	150.00	21,550.00	950.00
Short-Sleeve Classic Jersey- M	54.00	20,790.00	1,134.00
Short-Sleeve Classic Jersey- S	54.00	21,384.00	486.00
Short-Sleeve Classic Jersey- L	108.00	19,764.00	324.00
Short-Sleeve Classic Jersey- XL	0.00	21,168.00	918.00
Long-Sleeve Logo Jersey- S	0.00	20,350.00	1,100.00
Long-Sleeve Logo Jersey- XL	0.00	19,850.00	850.00

23. Calculate the quantities of each subcategory ordered in each month of 2013. Expected output as ahead:

Sub-Cat_Ordered	Jan #	Feb #	Mar #	Apr #	May #	Jun #	Jul #	Aug #	Sep #	Oct #	Nov #	Dec #
Bike Racks	13	21	35	18	25	26	20	25	27	28	38	31
Bike Stands	8	16	20	26	24	24	17	18	18	31	23	12
Bottles and Cages	270	491	567	563	602	766	672	759	711	777	728	811
Caps	62	149	160	149	163	198	183	215	193	200	192	235
----- 13 More Records Here -----												

24. Find the top-5 product names by average sales in each category for the year 2013 (if more than one product name in top-5 have same avg sales, include all of them). Expect output as formatted ahead:

category	same_rank_products	rank
Accessories	All-Purpose Bike Stand	1
Accessories	Hitch Rack - 4-Bike	2
Accessories	Hydration Pack - 70 oz.	3
Accessories	HL Mountain Tire	4
Accessories	Sport-100 Helmet- Red	5
Bikes	Road-250 Black- 44, Road-250 Black- 48, Road-250 Black- 52, Road-250 Black- 58, Road-250 Red- 58	1
Bikes	Touring-1000 Blue- 46, Touring-1000 Blue- 50, Touring-1000 Blue- 54, Touring-1000 Blue- 60, Touring-1000 Yellow- 46, Touring-1000 Yellow- 50, Touring-1000 Yellow- 54, Touring-1000 Yellow- 60	2
Bikes	Mountain-200 Silver- 38, Mountain-200 Silver- 42, Mountain-200 Silver- 46	3
Bikes	Mountain-200 Black- 38, Mountain-200 Black- 42, Mountain-200 Black- 46	4
Bikes	Road-350-W Yellow- 40, Road-350-W Yellow- 42, Road-350-W Yellow- 44, Road-350-W Yellow- 48	5
Clothing	Women's Mountain Shorts- L, Women's Mountain Shorts- M, Women's Mountain Shorts- S	1
Clothing	Classic Vest- S	2
Clothing	Classic Vest- L, Classic Vest- M	3
Clothing	Short-Sleeve Classic Jersey- L, Short-Sleeve Classic Jersey- M, Short-Sleeve Classic Jersey- S, Short-Sleeve Classic Jersey- XL	4
Clothing	Long-Sleeve Logo Jersey- XL	5

25. Generate the following business metrics (as formatted ahead):

Item	Value
Total Revenue	29,355,985.00 exact formatting (decimal precision included)
Average Price	486.04 rounding decimal to 2 digits
Total Units Sold	60423
Max Sales Order ID	SO46602 multiple order numbers (called 'order ID' here) may have max. sales value; pick highest order number
Min Sales Order ID	SO52187 multiple order numbers (called 'order ID' here) may have min. sales value; pick highest order number
Highest Sales Date	2011-12-28 latest date with highest sales
Lowest Sales Date	2013-02-03 latest date with lowest sales
Highest Sales Category	Bikes
Lowest Sales Category	Bottles and Cages, Cleaners, Helmets

many categories may have highest/lowest sales; return distinct category values for both

***** MISCELLANEOUS *****

- Create a function for calculating factorial of a number. Invoke it in the statement `SELECT your_factorial_func_name(30)` to find the factorial of, say, 30.
- Executing on the `products` table, `SELECT SUM('1+2') FROM products` yielded 295, `SELECT SUM(1+2) FROM products` yielded 885, and `SELECT SUM(5) FROM products` yielded the result 1475. Why were such outputs returned?
- A bar chart in a sales report of an imaginary car selling firm indicated that in Q1, Q2, Q3 and Q4, it sold \$100 million, \$180 million, \$220 million and \$240 million worth of cars, respectively. Is there anything to worry about?
- For each leap year between 1825 and 2025, write a SQL query to find the count of months in that year that had 13th on a Friday.
- How will you persist the results of a CTE for using the results in multiple `SELECT` statements in the same session?
- In a table having 1 billion rows spanning 100 columns of multiple data types, how do we decide which columns should be selected for creating indexes on?
- How exactly do the indexes on big tables speed-up the query processing?
- A product's new feature is meeting the end-users' needs but missing the business objectives. As a DA, what strategies would you recommend to address the imbalance?
- You've a column with values in order (top to bottom): Thu, Sat, Fri, Sun, Tue, Mon, Wed. Write a query to sort them chronologically ascending (Sun to Sat).
- A CSV of students' data has column MARKS which ideally should've positive INT values (from 0 to 100), but presently has a mix of values like 88.58, -35, 47, sixty three, fifty four, 92, 16.8, NA, two hundred six, zero, 0, ... etc. There are no special characters in the values. As evident, fifty four is acceptable & should be kept as 54, but two hundred six is to be obviously discarded/imputed/replaced. What tactics would standardize the MARKS column so that it contains legitimate INT values from 0 to 100?
- Pick any table having NULLs spread across its attributes over all the records. Write a query to return the number of NULLs per record in an adjoining column.
- How's the working of `SELECT COUNT(DISTINCT some_column)` different from `SELECT DISTINCT COUNT(some_column)`?
- A burger chain is thinking of opening a branch in Mumbai. What data would you pull-up to identify opportunities/risks so as to advise the management?
- As a data analyst, what challenges will you have to overcome in order to create an analytics report that is based on 3 big tables, one table hosted locally, one table in AWS, and one table in Snowflake Enterprise?
- If you were allowed to guess, how would you estimate the current total number of data analysts working in Bengaluru?
- A big data table is stored locally & its copy on cloud. A query on the table took 65 seconds locally, & ~4 seconds in cloud. What could be the reasons for this speed-up?
- When should one prefer the snowflake schema to the star schema? Describe a scenario.
- A data model is configured correctly & optimally with efficient DAX calculations in Power BI. The data set underlying the model is 100% cleaned and standardized. Still, the BI dashboard is not performant. What could be the reason(s)?
- An `EMPLOYEES` table with few million records has `DEPT` (TEXT, NOT NULL) and `SALARY` (DECIMAL, NOT NULL) columns (among other columns). `DEPT` has 5 distinct, non-NULL values. Write a SQL query to fetch the median salary of every department (i.e. the output should have only 5 rows & two columns as `DEPT` and `MEDIAN_SALARY`).
- What design tweaks would you implement if your BI dashboard's end user had achromatic vision?
- During query execution, aggregate functions run along with `GROUP BY` phase. A sample SQL query has `AVG(numeric_column)` in `SELECT` and also in `HAVING`. Which `AVG()` is executed first?
- You wrote a trigger on a table which, in turn, is the basis of your automated analytics query. The analytics query runs only after the trigger is fired post a table update. But today, that trigger didn't fire post table update. Thus, your analytics query didn't execute. Diagnose the trigger failure.
- Describe the various types of SCDs with examples.
- How do views, materialized views, temporary tables, CTEs and sub-queries differ from each other?
- A brick-&-mortar grocery store is migrating to online store for extended outreach & other benefits. What data-related aspects you reckon it will have to deal with?
- Describe some scenarios where `FULL OUTER JOIN` and `CROSS JOIN` would be needed.
- How do we decide the acceptable level of granularity in the information to be presented through charts/dashboards?
- What are type-1 and type-2 errors in hypothesis testing?
- You shared a Power BI report with the marketing team but one member of the team informed that she's not able to access the report. Possible reasons?
- A table `STUDENT` (blue table) has two columns: `STU_ID` (INT type) and `STU_INFO` (of JSON/NVARCHAR type in MySQL/T-SQL). Write query to generate results in red table.

STU_ID	STU_INFO
101	{"STU_NAME": "Alice", "AGE": 15, "MAJOR": "English"}
102	{"STU_NAME": "Bob", "AGE": 16, "MAJOR": "Physics"}
103	{"STU_NAME": "Charlie", "AGE": 17, "MAJOR": "Math"}
104	{"STU_NAME": "Diana", "AGE": 15, "MAJOR": "Math"}
105	{"STU_NAME": "Ethan", "AGE": 16, "MAJOR": "Physics"}
106	{"STU_NAME": "Fiona", "AGE": 17, "MAJOR": "English"}
107	{"STU_NAME": "George", "AGE": 15, "MAJOR": "Math"}
108	{"STU_NAME": "Hannah", "AGE": 16, "MAJOR": "Physics"}
109	{"STU_NAME": "Ian", "AGE": 17, "MAJOR": "Math"}
110	{"STU_NAME": "Jane", "AGE": 15, "MAJOR": "English"}
111	{"STU_NAME": "Tom", "AGE": 19, "MAJOR": "English"}

MAJOR	STU_COUNT	AVG_AGE	STU_IDs
English	4	16.5	101, 106, 110, 111
Math	4	15.75	103, 104, 107, 109
Physics	3	16.33	102, 105, 108